



🕒 Time: 5 minutes

🕒 Outcome: Calculate CPI and inflation using a market basket.

🕒 Difficulty: Intermediate

CPI and Inflation Measurement



THE CORE IDEA

The consumer price index compares the current cost of a fixed consumer basket with its base-year cost. The index equals the current basket cost divided by the base-year basket cost, multiplied by one hundred. Inflation is the percentage change in the index between periods. Holding basket quantities fixed isolates price change from changes in what consumers buy.



HOW TO RECOGNIZE IT

- Identify the fixed basket, its base-year cost, and its current-year cost.
- Compute the price index by comparing the current basket cost with the base-year basket cost.
- Measure inflation as the percentage change in the index from one period to the next.
- Keep basket quantities fixed when calculating the CPI for different years.



WATCH OUT

Do not use the wrong basket cost or forget to multiply by 100. Divide the current cost of the fixed basket by its base-year cost; the index can rise or fall as prices change.



WORKED EXAMPLE: CALCULATING CPI AND INFLATION

Fixed basket: quantities do not change

	Quantity	Base price	Current price	Base cost	Current cost
Item A	4	\$10	\$12	\$40	\$48
Item B	2	\$20	\$25	\$40	\$50
Basket total	—	—	—	\$80	\$98

The fixed basket costs \$80 in the base year and \$98 now. $CPI = (\$98 / \$80) \times 100 = 122.5$. Relative to the base-year index of 100, prices rose 22.5%. Separately, an index rising from 125 to 150 implies inflation of $(150 - 125) / 125 \times 100 = 20\%$.



CHECK YOURSELF

A basket costs \$400 in the base year and \$500 today. What is today's CPI?



READY?

Return to the game and master this concept.